

Section 11.6

1. 10 points If $x = x(y)$, $t = t(y)$ (that is, if both x and t represent functions of y), and $2\sqrt{x} = 11 + xt$, find $\frac{dt}{dy}$.

$$\begin{aligned}\frac{d}{dy}(2\sqrt{x}) &= \frac{d}{dy}(11 + xt) \\ 2\left(\frac{1}{2}(x)^{-1/2}\frac{dx}{dy}\right) &= \frac{d}{dy}(11 + xt) \quad \text{Chain Rule} \\ (x)^{-1/2}\frac{dx}{dy} &= \frac{dx}{dy}t + x\frac{dt}{dy} \quad \text{Product Rule} \\ (x)^{-1/2}\frac{dx}{dy} - \frac{dx}{dy}t &= x\frac{dt}{dy} \\ \frac{dx}{dy}((x)^{-1/2} - t) &= x\frac{dt}{dy} \\ \frac{dt}{dy} &= \frac{dx}{dy} \frac{((x)^{-1/2} - t)}{x}\end{aligned}$$

2. 10 points If $x = x(t)$, $y = y(t)$ (that is, if both x and y represent functions of t), and $3x^2 - 2y^3 = 11$, find $\frac{dx}{dt}$ when $x = 3$, $y = 2$, and $\frac{dy}{dt} = 6$.

$$\frac{d}{dt}(3x^2 - 2y^3) = \frac{d}{dt}(11)$$

$$3 \frac{d}{dt}(x^2) - 2 \frac{d}{dt}(y^3) = 0$$

$$3(2x \frac{dx}{dt}) - 2(3y^2 \frac{dy}{dt}) = 0$$

$$6x \frac{dx}{dt} - 6y^2 \frac{dy}{dt} = 0$$

$$6x \frac{dx}{dt} = 6y^2 \frac{dy}{dt}$$

$$\frac{dx}{dt} = \frac{6y^2}{6x} \frac{dy}{dt}$$

$$\frac{dx}{dt} = \frac{y^2}{x} \frac{dy}{dt}$$

$$\frac{dx}{dt} = \frac{(2)^2}{(3)}(6) \quad \text{Substitute values provided}$$

$$\frac{dx}{dt} = \frac{4}{3}(6) = 8$$

3. 20 points A manufacturing company estimates that weekly sales s and weekly marketing costs m (both in dollars) are related by

$$s = 45,000 - 10,000e^{-.005m}$$

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The current weekly marketing costs are \$200, and these are increasing at a rate of \$7 per week. Find the current rate of change of weekly sales.

$$s' = (45,000 - 10,000e^{-.005m})'$$

$$s' = -10,000(-0.005m')e^{-.005m}$$

$$s' = 50m'e^{-0.005m}$$

$$s' = 50(7)e^{-0.005(200)} \quad \text{Substitute values provided}$$

$$s' = 350e^{-1} = \frac{350}{e}$$

Section 11.7

1. 5 points Find the relative rate of change of $f(x) = 3 + 3e^{2x}$.

First, we find $f'(x)$ using the chain rule:

$$f'(x) = 6e^{2x}.$$

Then, we use the fact that the relative rate of change of a function f is given by $\frac{f'(x)}{f(x)}$ to yield a final answer of:

$$\frac{6e^{2x}}{3 + 3e^{2x}}.$$

2. 5 points Find the relative rate of change of $f(x) = 6x^3 - \ln(x)$ at $x = 3$.

First, we find $f'(x)$:

$$f'(x) = 18x^2 - \frac{1}{x}.$$

Then, we use the fact that the relative rate of change of a function f is given by $\frac{f'(x)}{f(x)}$ to yield:

$$\frac{18x^2 - \frac{1}{x}}{6x^3 - \ln(x)}.$$

Finally, we substitute $x = 3$ and reach a final answer of:

$$\frac{18(3)^2 - \frac{1}{(3)}}{6(3)^3 - \ln(3)} = \frac{162 - \frac{1}{3}}{162 - \ln(3)} = \frac{\frac{485}{3}}{162 - \ln(3)} = \frac{485}{3(162 - \ln(3))} \approx 1.0047561986.$$

3. 5 points Find the percentage rate of change of $f(x) = x^4 - 5$ at $x = 4$.

First ,we find $f'(x)$:

$$f'(x) = 4x^3.$$

Then, we use the fact that the percentage rate of change of a function f is given by $100 \left(\frac{f'(x)}{f(x)} \right)$ to yield:

$$100 \left(\frac{4x^3}{x^4 - 5} \right).$$

Finally, we substitute $x = 4$ to reach a final answer of:

$$100 \left(\frac{4(4)^3}{(4)^4 - 5} \right) = 100 \left(\frac{256}{256 - 5} \right) = 100 \left(\frac{256}{251} \right) \approx 105.577689243.$$

4. 10 points Given the price-demand equation $x = f(p) = 2000 - 3p^2$, find $E(p)$, the price-elasticity of demand. At $p = 15$, is demand elastic or inelastic?

Recall that the price elasticity of demand is given by the formula $E(p) = \frac{-pf'(p)}{f(p)}$. As such, we calculate $f'(p)$:

$$f'(p) = -6p.$$

The, we use the formula to yield our price-elasticity of demand:

$$E(p) = \frac{-pf'(p)}{f(p)} = \frac{-p(-6p)}{2000 - 3p^2} = \frac{6p^2}{2000 - 3p^2}.$$

To determine if demand is elastic or inelastic at $p = 15$, we substitute to yield:

$$E(15) = \frac{6(15)^2}{2000 - 3(15)^2} = \frac{1350}{1325} \approx 1.01886792453.$$

As demand is elastic when $E(p) > 1$, we may then conclude that the demand is elastic at $p = 15$.

5. Given the price-demand equation

$$3p^2 + 5x = 7500$$

- (a) 5 points Express the demand x as a function of the price p .

$$\begin{aligned} 3p^2 + 5x &= 7500 \\ 5x &= 7500 - 3p^2 \\ x &= \frac{1}{5}(7500 - 3p^2) \end{aligned}$$

- (b) 5 points Find the elasticity of demand, $E(p) = \frac{-pf'(p)}{f(p)}$.

First, we find $f'(p)$:

$$f'(p) = \frac{1}{5}(-6p) = \frac{-6p}{5}.$$

Then, we use the formula to calculate the elasticity of demand:

$$E(p) = \frac{-pf'(p)}{f(p)} = \frac{(-p)\frac{-6p}{5}}{\frac{7500-3p^2}{5}} = \left(\frac{6p^2}{5}\right) \left(\frac{5}{7500-3p^2}\right) = \frac{6p^2}{7500-3p^2}.$$

6. 10 points Given the price-demand equation $x = f(p) = \sqrt{200 - p^2}$, find the value(s) of the price p for which demand has unit elasticity.

First, we calculate $f'(p)$:

$$f'(p) = \frac{1}{2}(200 - p^2)^{-1/2}(-2p) = -p(200 - p^2)^{-1/2}.$$

Then, we calculate the elasticity of demand:

$$E(p) = \frac{-pf'(p)}{f(p)} = \frac{-p(-p(200 - p^2)^{-1/2})}{(200 - p^2)^{1/2}} = \frac{p^2}{(200 - p^2)^{1/2}(200 - p^2)^{1/2}} = \frac{p^2}{200 - p^2}.$$

Following this, we solve $E(p) = 1$ to reach our answer:

$$\begin{aligned} E(p) &= 1 \\ \frac{p^2}{200 - p^2} &= 1 \\ p^2 &= 200 - p^2 \\ 2p^2 &= 200 \\ p^2 &= 100 \\ p &= \pm 10. \end{aligned}$$

7. 10 points (bonus) Given the price-demand equation $x = f(p) = 10e^{-0.1p}$, what price p maximizes the revenue?

Revenue is equal to price times demand, so we construct the revenue function:

$$R(p) = pf(p) = p(10e^{-0.1p}) = 10pe^{-0.1p}.$$

Next, we calculate $R'(p)$:

$$R'(p) = 10(1)e^{-0.1p} + 10p(-0.1)e^{-0.1p} = (10 - p)e^{-0.1p}.$$

Setting $R'(p) = 0$, we solve to yield $p = 10$. Constructing a sign chart, we find that $R'(p) > 0$ on $(-\infty, 10)$ and that $R'(p) < 0$ on $(10, \infty)$. As such, at $p = 10$, the maximum revenue is achieved.